Practice workbook · When is a binomial expansion valid?

Stating the range of x for which a binomial series with a rational power converges.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-19

1. Question 7 · part b · 1 mark

2023 JUN · Paper 02

Worked-example retry · Attempt part b only.

7 (a) Expand $\left(1+\frac{x}{3}\right)^{-3}$ in ascending powers of x up to and including the term in x^3

Where appropriate express each coefficient as an exact fraction in its lowest terms.

(3)

(b) Write down the range of values of x for which your expression is valid.

(1)

(c) Express $(3+x)^{-3}$ in the form $P(1+Qx)^{-3}$ where P and Q are rational numbers whose values should be stated.

(2)

$$f(x)=\frac{(1+4x)}{(3+x)^3}$$

(d) Obtain a series expansion for $f(x)$ in ascending powers of x up to and including the term in x^2

(2)

(e) Hence, using algebraic integration, obtain an estimate of $\int_0^{0.2} f(x)\,dx$

Give your answer to 5 significant figures.

(3)

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1. Question 7 · part b · 1 mark · Working and answer

Working and answer

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2. Question 9 · part b · 1 mark

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

9 (a) Expand $(1 + 2x)^{-\frac{1}{3}}$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as a fraction in its lowest terms.

(3)

(b) Find the range of values of x for which your expansion is valid.

(1)

$$f(x) = \frac{2 + kx^2}{(1 + 2x)^{\frac{1}{3}}}$$

(c) Obtain a series expansion of f (x) in ascending powers of x up to and including the term in x^3
Give your coefficients in terms of k where appropriate.

(3)

The coefficient of x^3 in the series expansion of f (x) is $-\frac{8}{3}$

(d) Find the exact value of k

(2)

(e) Hence, using algebraic integration, estimate the value of

$$\int_{0.1}^{0.2} f(x)\, dx$$

Give your answer to 4 decimal places.

(5)

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2. Question 9 · part b · 1 mark · Working and answer

Working and answer

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3. Question 2 · part b · 1 mark

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

2 (a) Expand $\frac{2}{\sqrt{1+3x}}$ in ascending powers of x up to and including the term in x^3

Express each coefficient as a fraction in its simplest terms where appropriate.

(4)

(b) State the range of values of x for which the expansion is valid.

(1)

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3. Question 2 · part b · 1 mark

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

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Question 2 continued

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(Total for Question 2 is 5 marks)

3. Question 2 · part b · 1 mark · Working and answer

Working and answer

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